

Musi River Basin AI Flood Forecasting Report

This report summarizes AI-based rainfall-runoff forecasting, flood-risk classification, and predicted risk zonation for selected Musi River Basin locations.

Model Performance Metrics

Metric	Value
R ²	0.853
NSE	0.853
KGE	0.893
PBIAS (%)	-6.96
RSR	0.383
RMSE	21.13
MAE	16.32

Predicted Flood Risk Classification

Location	Rainfall	Discharge	Water Level	Risk
Vikarabad Upstream	82.51	134.03	2.09	Medium
Gandipet	42.52	78.52	1.57	Low
Himayat Sagar	20.53	66.56	1.52	Low
Uppal	66.79	128.96	2.1	Medium
Amberpet	45.49	93.34	1.78	Low
Chaderghat	75.35	135.97	2.1	Medium
Moosarambagh	82.22	143.43	2.05	Medium
Nagole	36.02	74.49	1.64	Low
Peerzadiguda	23.57	62.95	1.32	Low
Ghatkesar Downstream	111.72	177.43	2.49	High